Food Security Strategies: Building Resilience Against Natural Disasters

Stratégies de sécurité alimentaire : améliorer la résistance aux catastrophes naturelles

Strategien für die Sicherung der Ernährung: Stärkung der Widerstandsfähigkeit gegen Naturkatastrophen

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A large and growing share of the world's poor is exposed to a double risk of hunger and natural disasters. More than 800 million people are chronically undernourished and even their poor food security status is at risk because many of them live under conditions in which high risks of natural hazards coincide with high vulnerability. The number, intensity and costs of natural disasters are following a rising trend and pose particular challenges to food security. To address these challenges, a paradigm shift in policy strategy is recommended. Disaster risk management should be built *ex ante* into overall development policies and specifically into food security strategies. This article presents some suggestions as to how this could be realized.

Natural disasters – more frequent, more intense and more costly

Natural disasters are the product of two forces, hazard and vulnerability. They comprise both geo-physical events (earthquakes, tsunamis and volcanic eruptions) and hydro-meteorological hazards (landslides, droughts/famines, extreme temperatures, forest fires, floods, cyclones) as well as disease epidemics. In the first six years of this century, almost 400 natural disasters have been reported per year, twice the number of the period 1987–1998 (Figure 1). Part of the increase can be explained by better reporting. However, experts agree that a significant part is due to a growing frequency and intensity of disasters, mostly as a result of climate change (IPCC, 2007) and greater vulnerability of large sections of the world's population.

The higher vulnerability is mainly due to the fact that there are not only more and more people who live in hazard prone areas, including many mega-cities, but also growing values of physical capital are invested in such areas. (For a comprehensive global assessment of disaster risks see World Bank, 2005.) On average, 100,000 to 300,000 people per year are victims of disasters, including persons killed and persons affected. The average annual death toll was about 80,000 in the last decade. The annual global cost of disasters, mainly due to damage to houses, infrastructure and assets, has been in the region of US$ 70 billion in the last decade. Some experts predict that the costs may even reach yearly averages of US$ 300 billion in the next 50 years (Freeman *et al*., 2003, p.8).

Natural disasters hit the poor disproportionately

The effects of natural disasters are particularly adverse for the poor. Most developing countries are located in lower latitude regions that happen to be at far higher risk of natural hazards than those in the

In Nagapattinum, Tamil Nadu, India, exhausted fisherman who lost their homes, boats and livelihoods salvage what they can after the tsunamis ravaged the coast of India, Africa and Asia. *2005 © FAO/Ami Vitale.*

Northern hemisphere. Even though climate change is raising the frequency of extreme weather events in all world regions, the lower latitudes will be hit more. During the 1990s, more than 90 per cent of all major natural disasters occurred in developing countries. Half of the 49 Least Developed Countries, including most of the poor small island states, are classified as being at 'high-level disaster risk'. During the 1990s, 97 per cent of all deaths from natural disasters were in developing countries (Freeman *et al*., 2003). The costs relative to GDP are often significant. For example, during the 1990s the damage has been estimated at 16 per cent of GDP for Nicaragua and 2.5 percent for China (World Bank, 2004).

Also within countries the poor are hit disproportionately. Typically, due to poverty, race, class, gender or ethnicity, the poor live in houses that are ill-protected against destruction by earthquakes or wind storms, or they live in lowlands that are the first to be covered by floods.

About two thirds of the poor live in rural areas and depend on agriculture and healthy ecosystems for their livelihood. Many farm on fragile and dry lands without sufficient water storage and irrigation. Farming on these ecosystems is extremely vulnerable to natural hazards and the effects of climate change.

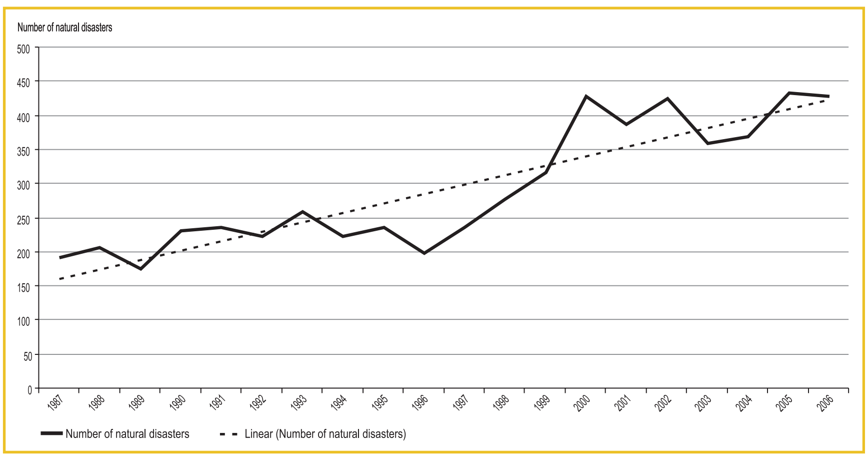
Close to Banda Aceh, Indonesia, this photo shows the devestation caused by the ingression of the sea into once fertile paddy lands, now covered with debris carried inland. *2005 © FAO/John Spaul.*

Finally, the poor are often discriminated against additionally in the aftermath of disasters, if public funds previously used for various social programmes are redirected in favour of compensation programmes for those who lost assets in the catastrophe. The poorest (e.g. homeless, tenants and women) have little or no assets for which they can claim compensation.

Policy responses

A paradigm shift is needed. The aim of policy response to disaster risk should be to reduce vulnerability and build resilience. To achieve this, a paradigm shift is needed in both development policy and disaster management strategies. For too long, both have been pursued in parallel. Even if the damaging impacts of disasters often could not have been avoided, they could have been reduced through pre-emptive investments. Some authors speak, therefore, of many disaster damages as 'manifestations of unresolved problems of development' (Yodmani, 2001). To avoid the possibility that natural disasters arrest past development efforts, resilience against disasters should be built as much as possible '*ex ante*' into development efforts in hazard prone locations, instead of repairing the damage '*ex post*' only (Twigg, 2004). It is particularly critical to integrate disaster risk management into food security strategies as part of overall poverty reduction and development policies.

Figure 1: Number of natural disasters, 1987–2006.



Source: Hoyois, Below, Scheuren and Guha-Sapir (2007).

A triple-track food security strategy for disaster prone areas. Almost 1 billion people live in absolute poverty. More than 800 million of them are chronically undernourished. Progress in fighting food insecurity has been stubbornly slow, although a number of countries have succeeded. Most successful countries have been following a 'twin-track food security strategy', combining investment in productivity growth for the poor, focussed especially on small-holder agriculture and rural infrastructure

Table 1: A triple-track approach to food security in disaster-prone countries

Track	Food Availability	Access to Food	Stability of Supplies	Food Utilisation
One Rural development and productivity enhancement	Investment in: Productivity, Natural resources, Infrastructure, Knowledge, Markets.	Promoting income opportunities: Access to assets, Creation of rural enterprises, Functioning of finance and labour markets.	Facilitating diversification and reducing instability: Reducing production variability (irrigation, water harvesting, pest control etc), Improving access to credit, Protecting natural resources.	Food handling and storage, Food safety regulation and institutions, Safe drinking water and sanitation.
Two Ensuring direct and immediate access to food	Food aid, Market information, Transport and communication.	School meals, Food for work programmes, Community and extended family structures.	Social safety nets.	Nutrition intervention, health and education programmes.
Three Building greater resilience against natural disasters	Risk information, analysis and early warning, Legislation, settlement and land use planning, Upgrading physical infrastructures, Diversification away from disaster risk, Strengthening risk transfer mechanisms (insurance and capital markets, contingency funds), Improving transition and sequencing of emergency rehabilitation-development efforts.			

(track One), with the creation of social safety nets to ensure direct and immediate access to food for the neediest (track Two). The details of such strategies have been elaborated in an Anti-Hunger Programme presented by FAO (FAO, 2003). Where the poor are exposed to a double risk of hunger and disaster, the twin-track approach should be expanded to include a 'third track' of measures that address the disaster vulnerability of people and their assets.

For various reasons, resilience building may in principle be needed in all four elements of food security – availability, access, stability and utilization – because disasters can affect all of them, resulting in more unstable food supplies, reduced access to food or disease and malnutrition. As agriculture is vulnerable to various disaster risks, measures to protect lands, water and assets are vital for limiting shortfalls in food security for the poor. Furthermore, poor farmers' practices of land use for agriculture, livestock, fisheries and forestry, including the management of related natural resources (such as water and biodiversity), need to be adapted to make them less vulnerable.

Some practical entry points

Some concrete entry points for interventions to build resilience against disasters for countries, communities or households are indicated in Table 1 as a third track of food security policies.

Risk information, analysis and early warning. Risk information and analysis encompasses hazard as well as vulnerability and resilience. It generates timely early warning and identifies potential protective and adaptive actions (see Figure 2). This should include the identification of successful examples, including indigenous knowledge. For example, in the case of the Asian tsunami, some ethnic groups, such as the

A child stands in front of a cow carcass near the village of Dasheq near Wajir, Kenya; most pastoralists lost nearly 90 per cent of their animals in the ongoing drought.
2005 © FAO/Ami Vitale.

Prolonged dry spells and erratic rainfall in Zimbabwe, Swaziland and Lesotho resulted in one of their worst main season harvests ever. *2007 © FAO photo.*

Moken of Thailand, abandoned their villages before the tsunami hit, thanks to traditional lore which prompted them to head to the hills when sea levels receded.

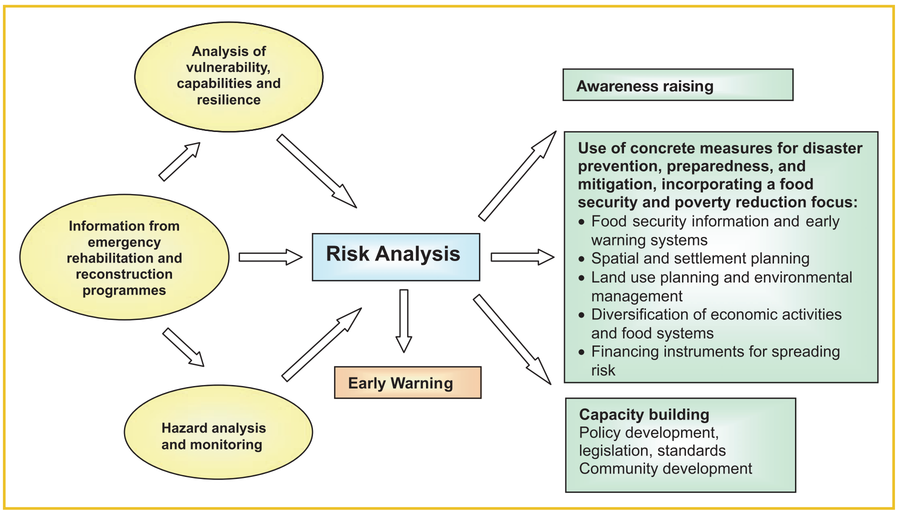
Legislation, settlement and land use planning. It is critically important that measures to enhance resilience become an integral part of settlement and land use planning policies. For example, some 50 to 70 per cent of residents in Bogota, Bombay, Delhi, Buenos Aires, Lusaka, Dar-es-Salaam and Kinshasa live in informal, often unsafe settlements (UNDP, 2004, p. 59). Hazard mapping, land-use regulation and codes for buildings and infrastructure as well as training of rural and urban developers need to ensure that settlements are made more disaster-resilient. In extremely risky locations, serious consideration must even be given to banning construction and to relocating populations.

Upgrading physical infrastructure. Four categories of investments in physical infrastructure are classic examples of disaster risk reduction measures: (1) adoption of standards and building codes that ensure adequate levels of robustness of houses and bridges for protection against earthquakes and hurricanes; (2) large scale engineering investments in dams, dykes to control floods, seawalls to break storm surges, rerouting rivers and building canals; (3) irrigation, water harvesting and water storage to prepare for drought; and (4) protection of forests and residential areas against large-scale fires.

Diversification away from disaster risk. Diversification of sources of livelihood and production is another well-established strategy of risk aversion. At least four levels can be distinguished: (1) livelihood diversification in the form of multiple sources of income is generally central to poor people's coping strategies; (2) trade liberalisation and the promotion of market links can reduce the risk of disruption of supplies from a limited source – for example, it has been shown that the liberalization of Bangladesh's rice imports from India prior to the 1998 floods prevented a steep hike in prices and thus a repetition of the famine that followed similar floods in the past (Del Ninno *et al*., 2003); (3) diversifying agricultural production can reduce vulnerability, e.g. the introduction of plants that are more resistant to wind, water, drought or salt water intrusion; (4) investments in water use efficiency, water storage and expanded irrigation systems.

Strengthening risk transfer mechanisms. No matter how much is invested in disaster prevention and mitigation, a significant risk of

Figure 2: Inputs and outputs of risk analysis



Source: Adapted from GTZ (2004).

> "Maßnahmen zur Erhöhung der Widerstandsfähigkeit gegen Katastrophen sollten in gefährdeten Gebieten integraler Bestandteil von Politikmaßnahmen und Strategien zur Sicherung der Ernährung sein ... ein Paradigmenwechsel muss her, und der aus diesem Ansatz entstehende Nutzen könnte die Kosten mehr als ausgleichen."

Sun-baked deposit of alluvial silt on the banks of the river which feeds the Shabbano Faraf reservoir, Iran. *1971 © FAO/H. Null.*

damage from natural disasters will remain in many hazard-prone areas. It is therefore appropriate that people and countries concerned have access to risk transfer mechanisms to cover the costs of rehabilitation. The two basic tools are insurance and capital market transactions (Banks, 2004). For a number of reasons, insurance faces limitations in developing countries: lack of sufficient risk capital; difficulties of re-insurance at country or regional levels due to high correlation of risk portfolio; inability of the poor to pay insurance premiums; unclear property rights. Developing countries need new forms of insurance, which entail lower costs, low premiums for the most needy and extension of re-insurance across a larger inter-regional space. Partly in response to the excessive costs of insurance systems, a second group of risk transfer mechanisms for developing countries has emerged in recent times. These are new financial instruments that link development finance with hedging against weather and natural disaster risks via capital markets. For example, 'catastrophe bonds' charge a high interest rate premium, but are subject to default if a defined catastrophe occurs during the life of the bond and 'weather derivatives' protect companies from climate variability under contracts that provide payouts in the event of a specified number of days with temperatures or rainfall above or below a specified trigger point.

In the absence of insurance and capital market instruments, the main financial tool used in developing countries is still contingent financing in the form of public calamity funds. The source of those funds can be national budgets as well as international grants or concessional loans. The advantage is that such funds are usually released quickly when a disaster occurs. However, contingency funding has two major shortcomings. One is the limited availability of public funds. The other is the moral hazard problem, sometimes referred to as 'Samaritan's dilemma' (Freeman *et al.*, 2003). Being aware and indeed expecting public assistance in the aftermath of a disaster, individuals, communities and the private sector in general have a limited incentive to take appropriate precautionary *ex ante* action.

> "Dans les zones exposées aux catastrophes naturelles, les mesures de renforcement de la résistance aux catastrophes devraient faire partie intégrale des politiques et des stratégies de sécurité alimentaire … il faut changer de paradigme et les avantages de cette approche pourraient de loin dépasser les coûts."

Benefits and costs

Evidently, integrating resilience building as a third track into food security strategies may be costly. However, it may also yield long-term benefits, which are too often neglected. Many governments of poor countries consider the extra costs of early warning, safer bridges, water storage or dams to lie beyond their capacity. Likewise, many of the private measures, e.g., reinforcing habitats or diversifying production systems are often, or are considered to be, unaffordable by those most at risk: the poorest. As a result, there is a tendency of under-investment in *ex ante* resilience building, and damage repair is mostly dealt with only *ex post* once a disaster has occurred, often relying on external help. This poses various challenges for action. One is cost reduction. Greater research efforts are needed to develop technical and institutional low-cost solutions to reduce vulnerability and enhance resilience. Another is raising awareness of the long-term benefits of *ex ante* measures. All concerned should be advised to decide on the basis of realistic benefit cost comparisons. Realistic benefit assessments must include at least tentative valuations of human lives to be saved and of irreparable losses of natural resources to be prevented. With the inclusion of long-term benefits, governments or individuals will presumably conclude in more cases that the perceived discounted present value of net benefits from *ex ante* investments in disaster

A subsistence farmer examines his banana crop destroyed by flooding in the Limpopo River Basin, Mozambique. *2000 © FAO/Clive Stanley.*

resilience exceeds the returns from 'normal' investments seeking mainly short-term profits. As a result, there would be more investment in resilience building, and disaster damage and the related costs of *ex post* rehabilitation would decline. In view of the long-term advantage of such integrated triple-track approaches, donors are well advised to support poor countries going in this direction and so are governments to assist poor individuals in such efforts.

In fact, some available estimates of costs and benefits often point to considerable returns. For example, it has been estimated that investments of US$ 40 billion in disaster preparedness, prevention, and mitigation would have reduced global economic losses in the 1990s by US$ 280 billion (Freeman *et al*., 2003).

A paradigm shift is needed

A large and growing share of the almost 1 billion poor live under the double risk of chronic undernourishment and exposure to natural disasters. A lot more can be done to improve preparedness, enhance resilience and protect past development investments against destruction. Rural development and food security policies and disaster management should be better integrated. In the past, food security policies have normally not been associated with 'direct life saving'. On the other hand, humanitarian assistance following disasters has not sufficiently sought to promote an 'escape from poverty' as part of the strategy (Twigg, 2004). A paradigm shift is needed, integrating the two concepts.

> "In disaster-prone locations measures to improve disaster resilience should be an integral part of food security policies and strategies ... a paradigm shift is needed and the benefits of this approach could significantly outweigh the costs."

The challenge is how to translate the proposed paradigm shift into practice. This will require political will and commitment by all stakeholders. Drawing lessons from practical examples, risk analyses and more realistic valuation of long-term benefits from *ex ante* investments in enhanced resilience are important. The enormous effort cannot be shouldered by the public sector alone. To engage the private sector more, various ways to overcome ignorance and create incentives need to be considered. To reduce potential moral hazard, legislation needs to be in place, including even obligatory community level contingency funds or mandatory insurance coverage.

The past decade has shown that public investments (national as well as international), have increasingly neglected rural sectors of developing countries, in spite of their key importance for food security, poverty alleviation and economic growth. Given the current nascent political will at all levels to step up such investment, the opportunity should be seized to build enhancement of disaster resilience into food security and agricultural development strategies.

A sunflower ready for harvest. *1997 © FAO/Giampiero Diana.*

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summary

Food Security Strategies: Building Resilience Against Natural Disasters

A large and growing share of the world's poor lives under a double risk. They are food insecure and live in conditions in which high risk of natural hazards coincides with high vulnerability. As a result natural disasters hit the poor disproportionately. This article argues that in order to mitigate disaster impacts on poor population groups, development policy and disaster management need to become more mutually supportive. Focusing on food security, it is suggested that in disaster-prone locations, measures to improve disaster resilience should be an integral part of food security policies and strategies. A 'triple track strategy' is recommended. In addition to investments in productivity enhancement, primarily for the rural poor (track one) and social safety nets for the most needy (track two), such a strategy would entail cross-cutting investments in building resilience against disaster impacts and risk management. Practical areas requiring more attention include risk information and analysis; land use planning; upgrading physical infrastructures; diversification and risk transfer mechanisms. As agriculture is particularly vulnerable to disaster risk, protecting agricultural lands, water and assets, should get greater weight in development strategies and food security policies. There is evidence that the benefits of this approach could significantly outweigh the costs.

Stratégies de sécurité alimentaire : améliorer la résistance aux catastrophes naturelles

Une part importante et croissante de la population mondiale est exposée à un risque double. Ces populations subissent l'insécurité alimentaire et vivent dans des conditions où le risque élevé d'aléas naturels va de pair avec une grande vulnérabilité. En conséquence, les catastrophes naturelles affectent les pauvres de manière disproportionnée. Cet article avance que, pour atténuer l'incidence des catastrophes sur les groupes de population défavorisées, la politique de développement et la gestion des catastrophes doivent se renforcer mutuellement. En mettant l'accent sur la sécurité alimentaire, il est suggéré que dans les zones exposées aux catastrophes naturelles, les mesures de renforcement de la résistance aux catastrophes devraient faire partie intégrale des politiques et des stratégies de sécurité alimentaire. Une "stratégie en trois axes" est recommandée. Outre des investissements pour augmenter la productivité visant principalement les populations rurales pauvres (axe 1) et des filets de sécurité sociale pour les plus nécessiteux (axe 2), une telle stratégie devrait comprendre des investissements non sectoriels pour affronter les conséquences des catastrophes et pour la gestion des risques. Les domaines concrets sur lesquels il faudrait se pencher plus attentivement comprennent l'information et l'analyse des risques; la planification de l'occupation des terres; la mise à niveau des infrastructures physiques; et les mécanismes de diversification et de transfert des risques. Comme l'agriculture est un secteur particulièrement vulnérable au risque de catastrophes, il faut accorder plus d'importance à la protection des terres, de l'eau et des actifs agricoles dans les stratégies de développement et les politiques de sécurité alimentaire. Il existe des éléments probants qui montrent que cette approche apporterait plus d'avantages que de coûts.

Strategien für die Sicherung der Ernährung: Stärkung der Widerstandsfähigkeit gegen Naturkatastrophen

Ein immer größerer Teil der Armen der Welt sieht sich einem doppelten Risiko ausgesetzt: Ihre Ernährung ist nicht gesichert, und ihre Lebensbedingungen weisen ein hohes Risiko an Naturgefahren bei gleichzeitig hoher Anfälligkeit auf. Daher werden arme Menschen unverhältnismäßig hart von Naturkatastrophen getroffen. In diesem Beitrag wird dargelegt, dass sich die Entwicklungspolitik und der Katastrophenschutz gegenseitig mehr unterstützen müssen, um die Auswirkungen von Naturkatastrophen auf die arme Bevölkerung abzumildern. Mit Blick auf die Sicherung der Ernährung wird vorgeschlagen, dass Maßnahmen zur Erhöhung der Widerstandsfähigkeit gegen Katastrophen in gefährdeten Gebieten integraler Bestandteil der Politikmaßnahmen und Strategien zur Sicherung der Ernährung sein sollten. Eine dreigleisige Strategie wird empfohlen: Neben den Investitionen zur Steigerung der Produktivität, die hauptsächlich den Armen im ländlichen Raum zu Gute kommen (Gleis 1), und in die sozialen Sicherheitsnetze für die Bedürftigsten (Gleis 2) würde eine solche Strategie übergreifende Investitionen in das Risikomanagement und die Widerstandsfähigkeit gegen die Auswirkungen von Naturkatastrophen mit sich bringen. Praxisnahe Bereiche, die mehr Beachtung erfordern, beinhalten Risikoinformation und –analyse, Flächennutzungsplanung, Ausbau der physischen Infrastruktur, Diversifikation sowie Risikotransfermechanismen. Da für die Landwirtschaft ein erhöhtes Katastrophenrisiko besteht, sollte dem Schutz von landwirtschaftlich genutzten Flächen, Wasser und Vermögen im Rahmen von Entwicklungsstrategien und Politikmaßnahmen zur Sicherung der Ernährung mehr Bedeutung beigemessen werden. Es gibt Anhaltspunkte dafür, dass der aus diesem Ansatz entstehende Nutzen die Kosten mehr als ausgleichen könnte.